

1) With regard to the NIST 800-131/2010 document, which are the recommended minimal levels of security for ES?

- a. Until 2010 – DSA : L=1024 bits; N=160 bits; RSA: K=1024 bits; ECDSA: f=160
- b. Until 2030 – DSA : L=4096 bits; N=256 bits; RSA: K=4096 bits; ECDSA: f=256
- c. Until 2010 – DSA : L=2048 bits; N=196 bits; RSA: K=2048 bits; ECDSA: f=196
- d. Until 2030 – DSA : L=2048 bits; N=224 bits; RSA: K=2048 bits; ECDSA: f=224

Let's break down NIST 800-131 and its implications for cryptographic key sizes, then figure out the correct answer. NIST SP 800-131A (the more commonly referenced version, even though you said 800-131/2010, which I assume means the initial version and the year) provides guidance on transitioning to stronger cryptographic algorithms and key lengths. Crucially, it provides *transitions*, not a single fixed minimum. The document defines timelines for phasing out weaker algorithms.

Here's a breakdown of how to tackle this, and what the correct answer's implications are:

- **DSA (Digital Signature Algorithm):** Uses two parameters, L and N. L is the length of the prime modulus, and N is the length of the cryptographic hash.
- **RSA:** Uses a key size (K) which represents the modulus length.
- **ECDSA (Elliptic Curve Digital Signature Algorithm):** Uses a field size (f), generally corresponding to the size of the curve's parameters, and therefore the key size.
- **Timelines:** The key concept in NIST 800-131A is the *transition*. There are dates after which certain key lengths are disallowed or deprecated.

Now, let's analyze the options:

- **a. Until 2010 – DSA : L=1024 bits; N=160 bits; RSA: K=1024 bits; ECDSA: f=160**  
This represents the *older*, weaker security level that NIST 800-131A was designed to *transition away from*. 1024-bit RSA and DSA (L=1024, N=160) were considered acceptable *before* the transition, but were being phased out.
- **b. Until 2030 – DSA : L=4096 bits; N=256 bits; RSA: K=4096 bits; ECDSA: f=256**  
This is too strong for the initial transition. 4096-bit keys are for much higher security levels, way beyond the *initial* minimum recommended by 800-131.
- **c. Until 2010 – DSA : L=2048 bits; N=196 bits; RSA: K=2048 bits; ECDSA: f=196**  
ECDSA and DSA values of 196 are not standard sizes. While there *might* be a specific, unusual configuration using 196, the standard sizes jump from 160-ish to 224+. This strongly suggests the answer is wrong.

- **d. Until 2030 – DSA : L=2048 bits; N=224 bits; RSA: K=2048 bits; ECDSA: f=224**  
This aligns with the transition recommendations within NIST 800-131A for a reasonable security level after the initial phase-out of 1024-bit keys. The "Until 2030" part is more in line with expectations and guidelines.

Therefore, the correct answer is **d**. The NIST guidelines provides transitions tables and option "d" respects those.

2) About public key Cryptographic Systems, we can say that they:

- a. Are symmetric
- b. Use different keys, that are math linked, for encrypting and decrypting
- c. Use different keys, randomly chosen, for encrypting and decrypting
- d. Are based on trap-door, one-way function

You are absolutely correct! I apologize for my previous, overly definitive response. In a multiple-choice question setting, sometimes more than one answer can be technically correct, and the best answer depends on the specific context and wording.

In this case, both options **b** and **d** are correct statements about public-key cryptographic systems:

- **b. Use different keys, that are mathematically linked, for encrypting and decrypting:** This accurately describes the fundamental *mechanism* of public-key cryptography. It highlights the asymmetric nature and the relationship between the public and private keys.
- **d. Are based on trap-door, one-way functions:** This accurately describes the underlying *mathematical principle* that enables public-key cryptography to work securely.

Therefore, a complete answer acknowledges that *both* **b** and **d** are correct. My previous response focused too narrowly on choosing a *single* "best" answer, which was inappropriate given the prompt's openness to multiple correct answers.

3) The RSA system uses:

- a. Operations with large integer number
- b. One key, publicly available
- c. One public key, one private key
- d. Operation on elliptic curves

Let's break down each option regarding the RSA cryptosystem:

- **a. Operations with large integer numbers:** This is **correct**. RSA's security relies on the mathematical difficulty of factoring very large integers (hundreds or thousands of bits long) into their prime factors. The core operations involve modular exponentiation with these large numbers.
- **b. One key, publicly available:** This is **incorrect**. RSA is a public-key cryptosystem, meaning it uses *two* keys: a public key and a private key. While the *public* key is publicly available, the *private* key must be kept secret. This statement describes a characteristic that *no* widely used, secure cryptographic algorithm uses.
- **c. One public key, one private key:** This is **correct**. This is the fundamental definition of a public-key cryptosystem, and RSA is a prime example.
- **d. Operation on elliptic curves:** This is **incorrect**. This describes Elliptic Curve Cryptography (ECC), a *different* type of public-key cryptosystem. RSA is based on integer factorization, not elliptic curves.

Therefore, the correct answers are **a** and **c**.

4) The revocation of certificate is done :

- a. When the public key inside the certificate is compromised
- b. When the validity of the certificate has expire
- c. By publishing the certificate according to the OCSP protocol
- d. When the private key is compromised

Let's examine each statement about certificate revocation:

- **a. When the public key inside the certificate is compromised:** This is **incorrect**. The *private* key being compromised is the primary reason for revocation. A public key, by its nature, is public. While there might be very rare, theoretical scenarios where a flaw is discovered in how a public key was generated (making it unexpectedly weak), this is not a common reason for revocation. The public key itself can't be "compromised" in the same way a secret can.
- **b. When the validity of the certificate has expired:** This is **incorrect**, but tricky. An expired certificate is *invalid*, but it's not the same as *revoked*. Expiration is a normal part of the certificate lifecycle. Revocation is for situations *before* the expiration date when the certificate is no longer trustworthy.

- **c. By publishing the certificate according to the OCSP protocol:** This is **incorrect**, and gets the mechanism backwards. OCSP (Online Certificate Status Protocol) is a way to *check* if a certificate has been revoked. It's a *method* for determining revocation status, not the act of revocation itself. The certificate is not published *according to* OCSP. A client uses OCSP to *query* a server about the status of a certificate.
- **d. When the private key is compromised:** This is **correct**. The most critical reason to revoke a certificate is if the corresponding private key is compromised (e.g., stolen, accidentally exposed). If an attacker has the private key, they can impersonate the certificate holder, decrypt sensitive information, or forge digital signatures.

Therefore, the best answer is **d**.

5) Which is the role of HSM (Hardware Security Module)?

- a. Keeps the used cryptographic algorithm secret
- b. Assures the protection of the cryptographic keys
- c. Accelerate the cryptographic operations
- d. Assures the protection of the computer which the CA software is installed

Let's examine the potential roles of an HSM (Hardware Security Module):

- **a. Keeps the used cryptographic algorithm secret:** This is **incorrect**. The cryptographic algorithms used (like RSA, AES, ECC) are generally *publicly known* standards. Security relies on the secrecy of the keys, *not* the algorithm. Keeping algorithms secret ("security through obscurity") is widely considered bad practice.
- **b. Assures the protection of the cryptographic keys:** This is **correct** and the primary purpose of an HSM. HSMs are designed with physical and logical security controls to prevent unauthorized access to the keys they store and manage. This includes tamper resistance, tamper evidence, and strong access controls.
- **c. Accelerate the cryptographic operations:** This is **correct**. Many HSMs include dedicated hardware to perform cryptographic operations (like encryption, decryption, signing, and key generation) much faster than a general-purpose CPU could. This is a significant benefit, especially for high-volume operations.

- **d. Assures the protection of the computer which the CA software is installed:**  
While an HSM *indirectly* contributes to the overall security of a system, this is not its primary role. A CA (Certificate Authority) system would *use* an HSM to protect its root and issuing keys, but the HSM itself doesn't protect the entire computer. Other security measures (firewalls, intrusion detection, operating system hardening, etc.) are responsible for protecting the broader computer system. The HSM is a component *within* a secure CA setup, not a replacement for general computer security.

Therefore, the best answers are **b** and **c**. B is the *primary* purpose, and C is a very significant, common feature.

#### 6) The certificate Policy –CP

- a. Describes the security which governs a CA
- b. Describes in detail the mechanism and procedures of the CA
- c. Describes the registration process of the users
- d. Represents an operational manual for the CA

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Let's analyze the options regarding a Certificate Policy (CP):

- **a. Describes the security which governs a CA:** This is a good, general description and is **correct**. A CP outlines the overall security framework, controls, and practices that a Certificate Authority (CA) adheres to. This includes things like physical security, personnel security, key management procedures, and audit requirements.
- **b. Describes in detail the mechanism and procedures of the CA:** This is also **correct**, but it's often more associated with a *Certification Practice Statement* (CPS) rather than a CP. While there's overlap, the CP sets the *high-level* policies, and the CPS often provides *more granular details* on *how* those policies are implemented. However, a CP *can* contain detailed mechanisms and procedures, so it's not entirely wrong.
- **c. Describes the registration process of the users:** This is **partially correct**, and a typical element *within* a CP (or its associated CPS). A CP will define the requirements and procedures for verifying the identity of entities requesting certificates (the registration process).
- **d. Represents an operational manual for the CA:** This is where it gets tricky. While a CP contains elements that inform operational procedures, it's not a full

"operational manual." An operational manual would be far more detailed and specific, covering day-to-day tasks and specific system configurations. The CP is more about *policy and requirements*, while an operational manual is about *implementation*. While the CP *informs* operations, it isn't the manual itself. It is too high-level to be considered a manual.

Therefore, **a**, **b**, and **c** are all correct aspects of a Certificate Policy. Option d is a distractor. It is too specific.

7) The law regarding the ES adopted by Parliament of Romania, in July 2001, establish :

- a. The fees for providing CA
- b. The legislative environment to electronic documents
- c. The conditions for providing electronic signatures certificate services
- d. The conditions for the ES of documents, in electronic communication

Let's break down what the Romanian Electronic Signature Law (likely referring to Law No. 455/2001) would likely establish:

- **a. The fees for providing CA:** While laws *might* touch on aspects of fees (e.g., maximum fees or fee structures), they typically don't *set* specific fees. Fees are usually determined by the service provider (the CA) based on market conditions and their own costs. So, this is unlikely to be a *primary* focus of the law.
- **b. The legislative environment to electronic documents:** This is **correct** and a core purpose of such a law. The law would establish the legal validity and recognition of electronic documents, provided they meet certain criteria (often related to the use of electronic signatures). This creates the framework for using electronic documents in place of paper documents.
- **c. The conditions for providing electronic signatures certificate services:** This is **correct**. The law would almost certainly define the requirements and regulations for entities acting as Certification Authorities (CAs) – the organizations that issue and manage digital certificates used for electronic signatures. This would include things like security requirements, audit procedures, and liability.
- **d. The conditions for the ES of documents, in electronic communication:** This is **correct** and is closely related to point (b). The law would define the conditions under which an electronic signature is considered legally valid and binding, providing a framework for secure electronic communication and transactions.

This might include specifying acceptable types of signatures, requirements for using secure devices, and so on.

Therefore, the correct answers are **b**, **c**, and **d**. The law establishes the *legal framework* for electronic documents, electronic signatures, and the services that provide them.

8) The extended electronic signature represents an electronic signature which complies with the following conditions:

- a. It is uniquely linked to the signer, assuring his identification
- b. Is created using specific hardware devices (smart cards or tokens)
- c. It is linked to the electronic document being signed
- d. It is created with means owned exclusively by signer

Let's analyze the conditions for what is often called an "advanced" or "qualified" electronic signature (the prompt uses "extended," which isn't a *standard* term, but implies a higher level of security than a basic electronic signature):

- **a. It is uniquely linked to the signer, assuring his identification:** This is **correct** and a fundamental requirement. The signature must be demonstrably tied to a specific individual, and there should be a robust process for verifying that identity.
- **b. Is created using specific hardware devices (smart cards or tokens):** This is **often a requirement**, but not *universally* so. Many regulations (like eIDAS in Europe) for *qualified* electronic signatures (the highest level) do require the use of a "qualified signature creation device" (QSCD), which is typically a smart card or token. However, "advanced" electronic signatures might have slightly less stringent requirements. It's more accurate to say this is *commonly* required for the highest assurance levels, but not *strictly* definitional of *all* "extended" signatures. So it *can* be correct.
- **c. It is linked to the electronic document being signed:** This is **correct** and essential. The signature must be inextricably linked to the data it signs. This prevents tampering – any change to the document after signing will invalidate the signature. This is usually achieved through cryptographic hashing.
- **d. It is created with means owned exclusively by signer:** This is **correct** and crucial for non-repudiation. The signing key (the private key) must be under the sole control of the signer. This means the signer, and only the signer, can create the signature.

Therefore, **a**, **c**, and **d** are definitely correct. **b** is also correct in most of the use cases (especially when the signature is considered "qualified"). So, all of them are acceptable answers.

9) RSA is a cryptographic algorithm based on:

- a. Difficulty of computing discrete logarithms in finite field
- b. Fermat's theorem
- c. Difficulty of computing exponents in finite field
- d. Difficulty of large integer numbers factorization

Let's examine the mathematical foundations of the RSA algorithm:

- **a. Difficulty of computing discrete logarithms in finite field:** This describes the basis of algorithms like Diffie-Hellman and ElGamal, *not* RSA.
- **b. Fermat's theorem:** While Fermat's Little Theorem *is* related to modular arithmetic and prime numbers, and *is used in the proof of correctness of RSA*, it's not the *core* hard problem RSA relies on. It's a necessary, but not sufficient, component.
- **c. Difficulty of computing exponents in finite field:** This is the opposite of the hard problem. Exponentiation in a finite field is relatively *easy* (and fast) to compute. RSA's security relies on the difficulty of the *inverse* operation.
- **d. Difficulty of large integer numbers factorization:** This is **correct**. RSA's security is based on the practical difficulty of factoring very large numbers (hundreds or thousands of bits) into their prime factors. Given a number  $n$  which is the product of two large primes  $p$  and  $q$  ( $n = p \cdot q$ ), it is computationally very difficult to find  $p$  and  $q$  knowing only  $n$ .

Therefore, the correct answer is **d**.

10) About time-stamping, we can say that::

- A. the RFC 3161 – Internet X.509 PKI Time Stamp Protocol (TSP), represents its standard environment
- B. is accomplished by a certificate service provider
- C. it's generated by the application and counter-signed by a time-stamping service provider
- D. the law on time-stamping, adopted in Romania, in 2004, represents its legislative environment

Let's analyze the statements about time-stamping:

- **A. the RFC 3161 – Internet X.509 PKI Time Stamp Protocol (TSP), represents its standard environment:** This is **correct**. RFC 3161 is the widely accepted standard for time-stamping protocols, defining how to request and receive trusted timestamps.
- **B. is accomplished by a certificate service provider:** This is **partially correct**, but needs clarification. While a *certificate* service provider (a CA) might also offer time-stamping services, the entity providing the time-stamping is more precisely called a *Time Stamping Authority* (TSA). A CA and TSA can be the same organization, but they have distinct roles. It's better to say a "Time Stamping Authority" or "Time-Stamping Service Provider."
- **C. it's generated by the application and counter-signed by a time-stamping service provider:** This is **correct**. The typical process involves:
  1. An application (or user) creates a hash of the data they want to time-stamp.
  2. This hash is sent to the Time Stamping Authority (TSA).
  3. The TSA combines the hash with the current time (from a trusted source) and digitally signs the result, creating a time-stamp token. This is effectively a "counter-signature" by the TSA.
- **D. the law on time-stamping, adopted in Romania, in 2004, represents its legislative environment:** This is **likely correct**, assuming Romania adopted a law on time-stamping in 2004 (which is plausible, as many countries did around that time to support e-commerce and digital signatures). Such a law would provide the legal framework for the recognition and validity of time-stamps within Romania. Without knowing the *specific* Romanian law number, we can say this is highly probable.

Therefore, **A**, **C**, and **D** are all likely correct. **B** is partially correct but imprecise.

11) Electronic document is equivalent, in condition of effect, with written documents if:

- a. It has an extended ES associated
- b. The signature is based on a qualified article
- c. The signature is created using a secure mechanism for creating digest signature
- d. The certificate service provider is accredited

Let's examine the conditions under which an electronic document can be considered legally equivalent to a traditional written document:

- **a. It has an extended ES associated:** This is **generally correct**, and is often phrased as "advanced electronic signature" or "qualified electronic signature." The higher level of assurance provided by such a signature (uniquely linked to the signer, under their sole control, and linked to the document) is crucial for legal equivalence. Using the term "extended" is not a problem, because, from the previous question, we know it is a synonym of advanced.
- **b. The signature is based on a qualified article:** This is **incorrectly worded** and unclear. It probably intends to say "qualified *certificate*." A "qualified certificate" is issued by a trusted Certificate Authority (CA) after rigorous identity verification, and it's often a requirement for the *highest* level of legal recognition of an electronic signature. So, while the *concept* is related to legal equivalence, the wording is flawed.
- **c. The signature is created using a secure mechanism for creating digest signature:** This is **partially correct, but incomplete and potentially misleading**. A "digest" (or hash) is *part* of the signature process. A secure hash function *is* necessary, but the statement focuses too narrowly on *just* the hashing aspect. It doesn't mention the private key, the certificate, or the overall signing process. It is a correct and mandatory condition, but not sufficient.
- **d. The certificate service provider is accredited:** This is **generally correct** and often a legal requirement. Accreditation (or a similar process like being a "trusted service provider" under eIDAS) means the CA meets specific standards for security, identity verification, and operational reliability. This provides confidence in the validity of the certificates they issue, and thus, the signatures that rely on those certificates.

Therefore, the best answers are **a** and **d**. "a" describes the type of signature needed, and "d" the requirement for the CA that support it. "b" is close but misworded, and "c" is too narrow.

12) The effort for breaking a symmetric system with a 128 bits key, with a specialized system, using brute force attack, is around:

- a.  $10^{18}$  years
- b. 3.6 hours
- c.  $18^{18}$  years
- d. 7000 years

Let's estimate the time required to break a 128-bit symmetric key via brute force.

- **Key Space:** A 128-bit key has  $2^{128}$  possible key combinations. This is an astronomically large number (approximately  $3.4 \times 10^{38}$ ).
- **Brute Force:** Brute force involves trying every possible key until the correct one is found. On average, you'd expect to find the key after searching half the keyspace ( $2^{127}$ ).
- **Specialized System:** We need to assume a very powerful, hypothetical "specialized system." Let's imagine a system that can test 1 trillion ( $10^{12}$ ) keys per second. This is far beyond current capabilities, but it gives us a benchmark for discussion.
- **Calculation:**
  - Keys to check:  $2^{127} \approx 1.7 \times 10^{38}$
  - Keys per second:  $10^{12}$
  - Seconds to check:  $(1.7 \times 10^{38}) / (10^{12}) = 1.7 \times 10^{26}$  seconds
  - Seconds in a year: Approximately 31,536,000 ( $3.15 \times 10^7$ )
  - Years to check:  $(1.7 \times 10^{26}) / (3.15 \times 10^7) \approx 5.4 \times 10^{18}$  years
- **Analyzing the Options:**
  - a.  $10^{18}$  years: This is in the right ballpark, given our very generous assumption about the "specialized system."
  - b. 3.6 hours: Completely unrealistic.
  - c.  $18^{18}$  years: This number doesn't make sense in this context. It is not a reasonable number.
  - d. 7000 years: Far too short.

Therefore, the closest (and still wildly optimistic) answer is **a.  $10^{18}$  years**. Even with an incredibly powerful, hypothetical system, cracking a 128-bit symmetric key via brute force is computationally infeasible within any practical timeframe. This demonstrates the strength of 128-bit symmetric encryption against brute-force attacks. The correct value is around  $5 \times 10^{18}$

13) DSA is a standard which:

- A. was created by Rivest-Shamir and Adleman
- B. is used for encrypting data
- C. can be implemented on elliptic curves
- D. is used only for digital signature

Let's evaluate the statements about the Digital Signature Algorithm (DSA):

- **A. was created by Rivest-Shamir and Adleman:** This is **incorrect**. Rivest, Shamir, and Adleman created the *RSA* algorithm. DSA was developed by the National Security Agency (NSA) and standardized by NIST.
- **B. is used for encrypting data:** This is **incorrect**. DSA is *exclusively* for digital signatures. It cannot be used for encryption.
- **C. can be implemented on elliptic curves:** This is **partially correct**, but requires nuance. The original DSA is based on the difficulty of the discrete logarithm problem in a finite field. However, there is a variant called *ECDSA* (Elliptic Curve Digital Signature Algorithm) which uses elliptic curve cryptography. So, while DSA *itself* is not elliptic curve-based, a closely related algorithm (ECDSA) is. The question does not mention a *variant* of DSA.
- **D. is used only for digital signature:** This is **correct**. This is the defining characteristic of DSA. Unlike RSA, which can be used for both encryption and digital signatures, DSA is *solely* for generating and verifying digital signatures.

Therefore, the best answer is **D**. Option C is a distractor, because can induce to error.

14) Diffie-Hellman key distribution scheme , with 3 users. The common values are the base,  $a=2$

and the modulus ,  $p=7$ . User A has  $\text{PubKXA} = 3$  and computes with  $\text{Priv KYA}$  . User B has  $\text{PubKXB} = 4$  and computes with  $\text{Priv KYb}$  . User C has  $\text{PubKXC} = 5$  and computes with  $\text{Priv KYc}$  .

The 3 users exchange their Public keys. Compute the secret keys used to encrypt the traffic

between users.

- a.  $KAB=1$  ,  $KAC=4$  ,  $KBC=1$
- b.  $KAB=4$  ,  $KAC=1$  ,  $KBC=1$

c.  $K_{AB}=4$  ,  $K_{AC}=1$  ,  $K_{BC}=4$

d.  $K_{AB}=1$  ,  $K_{AC}=1$  ,  $K_{BC}=4$

Let's break down the Diffie-Hellman key exchange and calculate the shared secrets.

## Understanding Diffie-Hellman

The core idea is:

1. **Public Parameters:** A prime number  $p$  (the modulus) and a base  $a$  (a primitive root modulo  $p$ ) are agreed upon publicly.
2. **Private Keys:** Each user chooses a secret private key (e.g.,  $K_{YA}$  for user A).
3. **Public Keys:** Each user calculates their public key:  $PubK_{XA} = a^{K_{YA}} \bmod p$  (e.g., for user A).
4. **Key Exchange:** Users exchange their *public* keys.
5. **Shared Secret Calculation:** Each user computes the shared secret using the *other user's public key* and their own *private key*:
  - User A calculates the shared secret with User B:  $K_{AB} = PubK_{XB}^{K_{YA}} \bmod p$
  - User B calculates the shared secret with User A:  $K_{BA} = PubK_{XA}^{K_{YB}} \bmod p$
  - Due to the properties of modular exponentiation,  $K_{AB} = K_{BA}$ .

## The Problem:

We have:

- $a = 2$
- $p = 7$
- $PubK_{XA} = 3$
- $PubK_{XB} = 4$
- $PubK_{XC} = 5$

We need to find the *private* keys ( $K_{YA}$ ,  $K_{YB}$ ,  $K_{YC}$ ) to calculate the shared secrets ( $K_{AB}$ ,  $K_{AC}$ ,  $K_{BC}$ ). We are provided with the public keys, not the private keys, so, they have to be computed before to compute the secret keys.

## Finding the Private Keys (Brute Force - Small Numbers)

Since  $p$  is very small (7), we can find the private keys by trying all possible values (from 1 to  $p-1 = 6$ ). We're looking for values that satisfy:

- $\text{PubKXA} = a^{\text{KYA}} \bmod p \Rightarrow 3 = 2^{\text{KYA}} \bmod 7$
- $\text{PubKXB} = a^{\text{KYB}} \bmod p \Rightarrow 4 = 2^{\text{KYB}} \bmod 7$
- $\text{PubKXC} = a^{\text{KYC}} \bmod p \Rightarrow 5 = 2^{\text{KYC}} \bmod 7$

Let's try the possibilities:

- **For KYA:**
  - $2^1 \bmod 7 = 2$
  - $2^2 \bmod 7 = 4$
  - $2^3 \bmod 7 = 1$
  - $2^4 \bmod 7 = 2$
  - $2^5 \bmod 7 = 4$
  - $2^6 \bmod 7 = 1$
- We do not find the requested value (3). The problem states that  $\text{PubKXA}=3$ , and so it does, but, because of  $p=7$  (too low for  $p$ ), it is not possible to compute KYA.  
**That is a problem in the exercise statement**

Let's find an approximate solution:

**Assuming the provided Public Keys are correct**, which is a mandatory requirement. The problem is well defined in that case.

**Calculating Shared Secrets** Because of the error in the problem definition, consider the given values:  $\text{PubKXA} = 3$   $\text{PubKXB} = 4$   $\text{PubKXC} = 5$

We'll use the provided *public* keys to compute what the shared secrets *would be*, if the private keys had been calculated correctly and resulted in those public key values. We cannot *verify* the consistency due to the problem with  $p=7$ , but we can perform the shared secret calculations *as if* the private keys were valid.

- **KAB (A and B):**
  - A would compute:  $\text{KAB} = \text{PubKXB}^{\text{KYA}} \bmod p$
  - B would compute:  $\text{KBA} = \text{PubKXA}^{\text{KYB}} \bmod p$
- **KAC (A and C):**
  - A would compute:  $\text{KAC} = \text{PubKXC}^{\text{KYA}} \bmod p$
  - C would compute:  $\text{KCA} = \text{PubKXA}^{\text{KYC}} \bmod p$
- **KBC (B and C):**

- B would compute:  $KBC = \text{PubKXC}^{KYB} \bmod p$
- C would compute:  $KCB = \text{PubKXB}^{KYC} \bmod p$

Because  $p=7$  and  $\text{PubKx}=3, 4$  or  $5$ , it does not allow to determine the values  $KYA$ ,  $KYB$  and  $KYC$ . Let's use trial-and-error using values, until a suitable condition is found. Consider  $KYA = 3$ ,  $KYB = 2$ ,  $KYC = 4$ . The values for  $\text{PubKX}$  should be computed.

$$\begin{aligned}\text{PubKXA} &= a^{KYA} \bmod p = 2^3 \bmod 7 = 1 \\ \text{PubKXB} &= a^{KYB} \bmod p = 2^2 \bmod 7 = 4 \\ \text{PubKXC} &= a^{KYC} \bmod p = 2^4 \bmod 7 = 2\end{aligned}$$

The assumption  $\text{PubKXA}=3$  and  $\text{PubKXC}=5$  does not hold.

$$\begin{aligned}\text{Let's try a combination, to calculate } KAB, KBC, KAC \\ KAB &= \text{PubKXB}^{KYA} \bmod p = 4^3 \bmod 7 = 1 \\ KAC &= \text{PubKXC}^{KYA} \bmod p = 5^3 \bmod 7 = 6 \\ KBC &= \text{PubKXC}^{KYB} \bmod p = 5^2 \bmod 7 = 4\end{aligned}$$

There is not solution within the provided ones. The correct answer must be based on  $KYA=3$ ;  $KYB=2$  and  $KYC=4$ , but the  $\text{PubK}$  values provided are incorrect, as already proved.

Using another set of keys, to match an option: Consider  $KYA = 5$ ,  $KYB = 2$ ,  $KYC = 4$

$$\begin{aligned}\text{PubKXA} &= a^{KYA} \bmod p = 2^5 \bmod 7 = 4 \\ \text{PubKXB} &= a^{KYB} \bmod p = 2^2 \bmod 7 = 4 \\ \text{PubKXC} &= a^{KYC} \bmod p = 2^4 \bmod 7 = 2\end{aligned}$$

$$\begin{aligned}KAB &= \text{PubKXB}^{KYA} \bmod p = 4^5 \bmod 7 = 4 \\ KAC &= \text{PubKXC}^{KYA} \bmod p = 2^5 \bmod 7 = 4 \\ KBC &= \text{PubKXC}^{KYB} \bmod p = 2^2 \bmod 7 = 4\end{aligned}$$

None of the answers is correct, using the values and the formulas. There is an error in the exercise statements. The key issue is that  $p=7$  is too small of a modulus for this to work properly.

**Looking at the options given that the stated values cannot have a result, it is not possible to have a solution**

With the values provided in the prompt, no valid DH exchange can occur that would produce these public key values with  $p=7$  and  $a=2$ .

Due the typo, I can not offer a correct solution.

The correct answer is marked as  $KAB=1$ ,  $KAC=1$ ,  $KBC=4$ . Can you analyze that?

Okay, let's analyze why the supposed "correct" answer ( $KAB=1$ ,  $KAC=1$ ,  $KBC=4$ ) *cannot* be achieved with the given parameters ( $a=2$ ,  $p=7$ ,  $\text{PubKXA}=3$ ,  $\text{PubKXB}=4$ ,  $\text{PubKXC}=5$ ) and standard Diffie-Hellman, and then discuss what *might* have been intended, or what error exists in the problem statement.

## Why the Given Answer is Impossible with the Stated Parameters

1. **Inconsistent Public Keys:** As we explored in the previous response, with  $a=2$  and  $p=7$ , the only possible public key values that can be generated are 1, 2, and 4. There's no way to generate public keys of 3 and 5 using valid private keys within this system. This is the fundamental flaw. The public keys given in the problem statement are *invalid* for the specified  $a$  and  $p$ .
2. **Let's Assume Valid Private Keys (Even Though They Don't Exist for the given PubKXs):** Even if we *pretend* we could find private keys  $KYA$ ,  $KYB$ , and  $KYC$  that *somehow* produced the given public keys (which we know is impossible), let's see if we could even *arrive* at the stated shared secrets. We'll do this by showing what the calculations *would* be, even though the starting point is flawed.
  - **To get  $KAB = 1$ :**
    - A would compute:  $KAB = \text{PubKXB}^{KYA} \bmod p = 4^{KYA} \bmod 7$
    - B would compute:  $KBA = \text{PubKXA}^{KYB} \bmod p = 3^{KYB} \bmod 7$
    - For this to equal 1, we'd need  $4^{KYA} \bmod 7 = 1$  AND  $3^{KYB} \bmod 7 = 1$ . We know  $4^3 \equiv 1 \bmod 7$ . But, as said before, A does not have  $\text{PubKey} = 1$
  - **To get  $KAC = 1$ :**
    - A would compute:  $KAC = \text{PubKXC}^{KYA} \bmod p = 5^{KYA} \bmod 7$
    - C would compute:  $KCA = \text{PubKXA}^{KYC} \bmod p = 3^{KYC} \bmod 7$
    - For this to equal 1, we'd need  $5^{KYA} \bmod 7 = 1$  AND  $3^{KYC} \bmod 7 = 1$ . It must be  $5^6 \bmod 7 = 1$ , but we did not find the private key
  - **To get  $KBC = 4$ :**
    - B would compute:  $KBC = \text{PubKXC}^{KYB} \bmod p = 5^{KYB} \bmod 7$
    - C would compute:  $KCB = \text{PubKXB}^{KYC} \bmod p = 4^{KYC} \bmod 7$
    - For this to be 4, we would need  $5^{KYB} \bmod 7 = 4$  AND  $4^{KYC} \bmod 7 = 4$ . It must be  $5^2 \bmod 7 = 4$ , and  $4^2 \bmod 7 = 2$  and  $4^4 \bmod 7 = 4$ , so  $KYB=2$  is ok, but  $KYC = 4$  is not.
3. **The core problem remains:** There are *no* integer values for  $KYA$ ,  $KYB$ , and  $KYC$  that will satisfy the equations with  $a=2$ ,  $p=7$ , and the *given* public keys (3, 4, and 5). The public keys are the root of the problem.

## Possible Explanations/Errors in the Problem

1. **Incorrect Public Keys:** The most likely error is that the public keys (PubKXA, PubKXB, PubKXC) were chosen incorrectly for the given  $a$  and  $p$ . Someone might have intended to use different values, or they might have made a mistake in calculating them.
2. **Different  $a$  or  $p$ :** It's *possible* (though less likely) that the intended problem used a different base ( $a$ ) or modulus ( $p$ ). If those values were different, then different public keys and shared secrets would be possible.
3. **Conceptual misunderstanding:** The person creating the question might have fundamentally misunderstood how Diffie-Hellman works.
4. **Typos:** There was a typo and some numbers were mixed up.

**In summary:** The "correct" answer provided is *mathematically impossible* given the other parameters in the problem. The problem itself is flawed, most likely due to incorrectly specified public keys. There's no valid Diffie-Hellman key exchange that can produce the provided public keys with  $a=2$  and  $p=7$ , making it impossible to arrive at *any* consistent set of shared secrets.

But what is the closest incorrect answer? Because professors do not care what is right or wrong.

You're right to consider the "professor's perspective" – sometimes the goal is to find the *least wrong* answer, even if the problem itself is flawed. Given the constraints and the impossible nature of the perfectly correct solution, let's re-evaluate the options to find the "closest incorrect" answer, assuming we *must* use the given (incorrect) public keys and that we're looking for *some* kind of pattern, even if it's not mathematically sound Diffie-Hellman.

Here are the options again:

a.  $K_{AB}=1$  ,  $K_{AC}=4$  ,  $K_{BC}=1$  b.  $K_{AB}=4$  ,  $K_{AC}=1$  ,  $K_{BC}=1$  c.  $K_{AB}=4$  ,  $K_{AC}=1$  ,  $K_{BC}=4$  d.  $K_{AB}=1$  ,  $K_{AC}=1$  ,  $K_{BC}=4$

Here's a breakdown of why we might choose option **d** as the *least wrong*, even though it's still based on a flawed premise:

1. **Focus on Patterns, Not Correct DH:** We're abandoning the idea of a *correct* Diffie-Hellman calculation because it's impossible with the given values. We have to look for *some* pattern or logic the professor *might* have intended, even if it's flawed.
2. **Option d:  $K_{AB}=1$  ,  $K_{AC}=1$  ,  $K_{BC}=4$**

- There could have been an attempt to get the value '1', using values closed to the powers of 2:  $2^3=8=1 \bmod 7$

### 3. Why other options are *even more* wrong:

- Options 'a' and 'b' have values mixed.

**In conclusion, although any solution is incorrect, Option d,  $KAB=1$ ,  $KAC=1$ ,  $KBC=4$ , is the best attempt.** It follows *some* logic.

It's crucial to understand that this is still *wrong* in terms of correct Diffie-Hellman. We're simply trying to reverse-engineer a flawed question to find the "least wrong" answer based on a likely misunderstanding or typo. If possible, it would be best to point out the error in the problem to the professor.

15) Which solution(s) was/were proposed for satisfying the WYSIWYG request?

(what you see is what you get)

- A. the signing of a graphic representation of the document
- B. the use of a secure display program, for viewing the documents
- C. the use of logos for certifying the authenticity of the document
- D. the use of a document format, which has an unique graphic representation

Let's break down the WYSIWYG (What You See Is What You Get) concept in the context of digital signatures and analyze the proposed solutions:

- **WYSIWYG in Digital Signatures:** The core problem is ensuring that the user *knows exactly what they are signing*. If a document can be displayed differently on different systems or with different software, the user might sign something that *looks* one way to them but is *actually* interpreted differently by the recipient (or a court). This is a major security and legal issue.

Now, let's analyze the options:

- **A. the signing of a graphic representation of the document:** This is a **correct** solution. If you convert the document into a fixed, unambiguous graphical format (like an image or a PDF/A, where PDF/A is specifically designed for long-term archiving and rendering consistency), and then sign *that* graphical representation, you ensure that what the user sees and signs is exactly what will be verified later. The signature is tied to the *visual representation*, not just the underlying data, which might be interpreted differently.

- **B. the use of a secure display program, for viewing the documents:** This is **partially correct, but insufficient on its own**. A "secure display program" *could* help ensure consistent rendering, *but* it relies on *everyone* using that same program (or a compatible one). This is a significant practical hurdle. It also doesn't address the issue of potential intentional manipulation *within* that program. It's a good *component* of a solution, but not a complete solution by itself.
- **C. the use of logos for certifying the authenticity of the document:** This is **incorrect**. Logos can be easily copied and forged. They provide no cryptographic guarantee of integrity or consistent rendering. They're a visual indicator, not a security mechanism.
- **D. the use of a document format, which has an unique graphic representation:** This is **correct** and related to option A. Certain document formats are designed to have a consistent visual representation across different platforms and software. PDF/A is the prime example. By using a format like PDF/A, you minimize the chances of the document being rendered differently, ensuring that what the signer sees is what others will see. This is about choosing a file format specifically designed for this purpose.

Therefore, the best answers are **A** and **D**. They address the core issue of ensuring a consistent, unambiguous visual representation of the document being signed. Option B is a helpful, but not complete solution, and option C is irrelevant to WYSIWYG.

16) If, at the moment of validating a signature, the signing certificate is revoked, the signature may be declared valid only if:

- A. the signed document was time-stamped after being stamped and before the revocation of the certificate
- B. the signed document was time-stamped before being signed
- C. the signature will be declared invalid anyway, because the certificate is revoked when the validations is done
- D. the certificate of the signer hasn't expired at the moment of validation

Let's analyze the situation of validating a digital signature when the signing certificate has been revoked:

- **The Core Problem:** A revoked certificate means the associated private key might have been compromised, or the certificate was otherwise deemed untrustworthy *before* its natural expiration date. Therefore, relying on signatures created *after* the revocation is generally a bad idea.

- **Let's look at the options:**

- **A. the signed document was time-stamped after being stamped and before the revocation of the certificate:** This is **incorrect**. If the document were time-stamped *after* being signed (and the timestamp is from a trusted Time Stamping Authority - TSA), and *that timestamp is before the certificate revocation*, then the signature *can* be considered valid. The critical point is that the *time-stamp proves* the signature existed *before* the certificate was revoked. The question states "stamped after being stamped", which is incorrect.
- **B. the signed document was time-stamped before being signed:** This is **incorrect**. If the document were time-stamped *before* it was signed, the time-stamp would be meaningless in proving the signature's validity. The time-stamp needs to be applied to the *signed* document (or, more precisely, a hash of the signed document).
- **C. the signature will be declared invalid anyway, because the certificate is revoked when the validations is done:** This is **generally the default outcome, but not always true**. It's the *safest* assumption. However, as explained above, a trusted time-stamp can provide an exception.
- **D. the certificate of the signer hasn't expired at the moment of validation:** This is **incorrect** and confuses revocation with expiration. A certificate can be revoked *before* it expires. Expiration is a normal part of the certificate lifecycle. Revocation is an exceptional event indicating a problem.

- **The Correct Principle (and the corrected A):** The key is a trusted time-stamp. If you can *prove*, using a time-stamp from a trusted TSA, that the signature was created *before* the certificate was revoked, then the signature can *still* be considered valid, *even though* the certificate is now revoked. This is because, at the time of signing, the certificate *was* valid.

The correct option should read as: **the signed document was time-stamped after being signed and before the revocation of the certificate** The prompted answers has typos.

So, none of the proposed options is correct.

Same thing, find the least wrong.

Okay, we're in "least wrong" mode again, trying to interpret what the question *intended* to ask, despite the errors. Let's re-examine the options, looking for the one that's closest to a correct concept, even if poorly worded:

- **A. the signed document was time-stamped after being stamped and before the revocation of the certificate:** This option is the *closest* to being correct, despite the nonsensical "stamped after being stamped" part. The *intent* is clearly about time-stamping, and the crucial phrase "before the revocation of the certificate" is present. This is the key to making an exception to the general rule of invalidating signatures from revoked certificates. The core idea is right, even with the typo.
- **B. the signed document was time-stamped before being signed:** This is completely illogical. A time-stamp applied *before* signing has no bearing on the validity of the signature.
- **C. the signature will be declared invalid anyway, because the certificate is revoked when the validations is done:** This is the *default* rule, and it's the *safe* assumption. But it's not *always* true, as the time-stamping exception exists. So, while it's a common outcome, it's not the *least wrong* answer here, because we're looking for something that hints at the exception.
- **D. the certificate of the signer hasn't expired at the moment of validation:** This is irrelevant. Expiration is distinct from revocation.

### Conclusion: The "Least Wrong" Answer

The "least wrong" answer is **A**, despite the "stamped after being stamped" error. We're choosing it because:

1. **Time-Stamping:** It correctly identifies time-stamping as the relevant mechanism.
2. **"Before Revocation":** It explicitly mentions the crucial timing requirement – the time-stamp must be *before* the revocation.

We're essentially interpreting the question as *trying* to ask about the time-stamping exception, even though it gets the details wrong. We are assuming the professor is testing the understanding about time stamps.

17) The secured mechanism for creating electronic signatures must satisfy the following conditions:

A. the signature creation data must be protected effectively by the signer against their use by unauthorized persons

B. it must not modify the electronic document which will be signed and not prevent its view by the signer, before finishing the signing process

C. the signature creation data must appear only once and its confidentiality must be assured

D. the signature creation data must not be deduced and the signature must be protected against forgery

Let's analyze the requirements for a secure electronic signature creation mechanism:

- **A. the signature creation data must be protected effectively by the signer against their use by unauthorized persons:** This is **correct**. The "signature creation data" refers primarily to the *private key*. Protecting the private key from unauthorized access is absolutely fundamental to the security of the entire system. If someone else gets your private key, they can forge your signature.
- **B. it must not modify the electronic document which will be signed and not prevent its view by the signer, before finishing the signing process:** This is **correct** and addresses the WYSIWYG (What You See Is What You Get) principle. The signing process shouldn't alter the document being signed, and the signer must be able to see the document *before* signing it to ensure they're signing what they intend to sign.
- **C. the signature creation data must appear only once and its confidentiality must be assured:** This is **partially correct, but the wording is slightly off**. The *private key* (the core of the signature creation data) should ideally be *generated* only once and kept strictly confidential. The phrase "appear only once" is a bit vague. It's not about how many times it *appears* (it might be stored securely in multiple backups, for example), but about ensuring it's only *generated* once to minimize the risk of compromise. The crucial part is the *confidentiality* of the private key.
- **D. the signature creation data must not be deduced and the signature must be protected against forgery:** This is **correct**. The private key (signature creation data) must be computationally infeasible to derive from the public key or any other publicly available information. Furthermore, the signature scheme itself must be resistant to forgery – it should be impossible for someone to create a valid signature without knowing the private key. This relies on the underlying cryptographic algorithms being strong.

Therefore, **A**, **B**, and **D** are definitively correct. **C** is mostly correct, emphasizing the critical need for confidentiality of the private key, but uses slightly imprecise wording. All of the provided options are acceptable.

18) Which of the following devices are specific for Secure Signature Creation Devices (SSCD)?

- A. Smart cards
- B. USB tokens
- C. PCMCIA
- D. Biometric readers

Let's examine the provided options in the context of Secure Signature Creation Devices (SSCDs):

- **SSCD Definition:** An SSCD is a hardware device specifically designed to securely store a private key and perform digital signature operations. Key requirements include tamper resistance, secure key generation, and protection against unauthorized access.
- **A. Smart cards:** This is **correct**. Smart cards are a classic example of SSCDs. They contain a small embedded chip with a secure microcontroller that can store private keys and perform cryptographic operations. They are physically tamper-resistant.
- **B. USB tokens:** This is **correct**. Many USB tokens are designed as SSCDs. They offer similar functionality to smart cards (secure key storage and cryptographic processing) but use a USB interface for connection.
- **C. PCMCIA:** This is **incorrect**, although it *could* contain an SSCD. PCMCIA (Personal Computer Memory Card International Association) is an older interface standard (like a larger version of a modern SD card slot) for connecting peripherals to laptops. A PCMCIA *card* could *potentially* contain an SSCD, but the PCMCIA interface *itself* is not an SSCD. It's like saying "a computer" is an SSCD – a computer *could* contain an HSM (a type of SSCD), but the computer itself isn't one.
- **D. Biometric readers:** This is **incorrect**. Biometric readers (fingerprint scanners, etc.) are used for *authentication* (proving *who* you are), but they don't *store private keys* or *perform signing operations*. They might be used as *part* of a multi-factor authentication system to *unlock* an SSCD, but they are not SSCDs themselves.

Therefore, the correct answers are **A** and **B**. Smart cards and USB tokens (specifically those designed for secure key storage and signing) are common examples of SSCDs.

19) The electronic signature assures:

- A. the authentication of the data link between sender and receiver
- B. the authentication of the signed message (its integrity verification)
- C. the authentication of the sender of the signed message
- D. the authentication of the receiver of the signed message

Let's analyze what an electronic signature (digital signature) assures:

- **A. the authentication of the data link between sender and receiver:** This is **incorrect**. Digital signatures don't directly authenticate the *communication channel* itself. Technologies like TLS/SSL (used in HTTPS) handle secure communication channels. The signature is on the *data*, not the *link*.
- **B. the authentication of the signed message (its integrity verification):** This is **correct**. A digital signature provides *integrity*. If the message is altered in any way after it's signed, the signature verification will fail. This assures the recipient that the message hasn't been tampered with.
- **C. the authentication of the sender of the signed message:** This is **correct**. A digital signature provides *authentication of origin* (also called non-repudiation). Because the signature is created using the sender's *private key*, and only the sender should possess that key, a valid signature proves the message came from that specific sender.
- **D. the authentication of the receiver of the signed message:** This is **incorrect**. Digital signatures have nothing to do with authenticating the *recipient*. Anyone with the public key can verify the signature; there's no restriction or authentication of who can *receive* the message.

Therefore, the correct answers are **B** and **C**. A digital signature provides both *integrity* (assurance the message hasn't changed) and *authentication of origin* (assurance of who sent the message).

20) The OCSP server, in order to obtain revocation information, may:

- A. process CRLs, issued by the Certification Authority
- B. call the services of another OCSP server, authoritative for the PKI domain in which the certificate was issued
- C. use SSL protocol
- D. use LDAP v2 protocol

Let's examine how an OCSP (Online Certificate Status Protocol) server obtains revocation information:

- **OCSP's Purpose:** An OCSP server provides real-time information about the revocation status of digital certificates. Clients can query the OCSP server to check if a certificate is still valid (not revoked) before relying on it.
- **A. process CRLs, issued by the Certification Authority:** This is **correct**. Certificate Revocation Lists (CRLs) are lists of revoked certificates published by a CA. An OCSP server can obtain revocation information by downloading and processing these CRLs. This is a common method.
- **B. call the services of another OCSP server, authoritative for the PKI domain in which the certificate was issued:** This is **correct**. OCSP servers can be configured in a hierarchical or delegated manner. If an OCSP server doesn't have information about a particular certificate, it might query another OCSP server that is authoritative for the CA that issued the certificate. This is called "OCSP stapling" or "OCSP delegation".
- **C. use SSL protocol:** This is **partially correct, but not directly related to obtaining revocation information**. OCSP requests and responses are often transmitted over HTTPS (which uses SSL/TLS) for *security* (to protect the integrity and confidentiality of the communication). However, SSL/TLS is the *transport* mechanism, not the source of the revocation data itself. The OCSP server doesn't "use SSL" to *get* the revocation list; it uses SSL to *communicate* securely.
- **D. use LDAP v2 protocol:** This is **potentially correct, but less common than CRLs**. LDAP (Lightweight Directory Access Protocol) *can* be used to access directory services that store certificate and revocation information. However, CRLs are the more standard and widely used mechanism for distributing revocation information. While a CA *might* publish revocation information via LDAP, it's not the primary method for OCSP.

Therefore, the best answers are **A** and **B**. CRLs are the most common and direct source of revocation information, and OCSP servers can delegate requests to other authoritative servers. C is about secure communication, not the *source* of data, and D is less common than A.

21) The law no. 589/2004, regarding the legislative environment for the electronic notary activity states that:

A. the electronic notary closure contains as signatures: the extended electronic signature of the parties, the extended electronic signature of the public notary, the extended electronic signature of the notary bureau

B. the electronic notary closure contains as signatures: the extended electronic signature of the parties, the extended electronic signature of the notary bureau

C. the public notary is obliged to keep the electronic archive of electronic notary documents for 20 years

D. the requests for an electronic notary document shall be handed to the public notary in electronic form, signed with the extended electronic signature of the requester

Let's analyze the statements about Romanian Law No. 589/2004 (assuming this law deals with electronic notary activities as the prompt suggests) in the context of electronic notary closures:

- **It's important to note:** Without the exact text of Law No. 589/2004, we're making educated guesses based on common practices in e-notary legislation and digital signature principles. However, laws vary, so this analysis is based on general principles and best practices, not specific legal knowledge of that particular Romanian law.
- **A. the electronic notary closure contains as signatures: the extended electronic signature of the parties, the extended electronic signature of the public notary, the extended electronic signature of the notary bureau:** This is **plausible and likely correct**. The core idea of an electronic notary closure is to have a legally binding digital record of the notarization.
  - *Parties' Signatures:* Essential for demonstrating their agreement to the document's contents.
  - *Public Notary's Signature:* Essential for the notarization itself, attesting to the verification of identities and the act of witnessing.
  - *Notary Bureau's Signature:* This is *possible* but less common. Some jurisdictions might require the notary *bureau* (the organization) to also have a digital signature, adding an extra layer of organizational accountability. It's less critical than the notary's individual signature, but not unreasonable.
- **B. the electronic notary closure contains as signatures: the extended electronic signature of the parties, the extended electronic signature of the notary bureau:** This is **less likely to be fully correct**. While the parties' signatures are essential, and the notary bureau's signature *might* be present, the *individual notary's* signature is the core of the notarization. Omitting the notary's personal signature would be highly unusual and undermine the legal validity.

- **C. the public notary is obliged to keep the electronic archive of electronic notary documents for 20 years:** This is **plausible**. Retention periods for legal documents vary by jurisdiction and document type, but 20 years is a reasonable timeframe for important legal records. This is a common requirement in many notary laws.
- **D. the requests for an electronic notary document shall be handed to the public notary in electronic form, signed with the extended electronic signature of the requester:** This is **plausible and likely correct**. To maintain a secure and verifiable digital workflow, requiring requests for e-notarization to be submitted electronically and signed with an extended (advanced/qualified) electronic signature makes perfect sense. This provides authentication of the requester and integrity of the request.

### **Best Guesses (Without the Specific Law Text):**

Based on common practices, **A**, **C**, and **D** are the most likely to be correct, or close to correct. A is preferred because of Notary Public signature.

22) Let there be a RSA protocol for transporting the key, with 2 users. User A publishes his public key [ $e_A = 3$ ,  $n_A = 55$ ] and computes his private key [ $d_A = 27$ ,  $n_A = 55$ ]. Let us suppose that user B wants to issue an encrypted message towards user A. User B envelopes the session key ( $K_{AB} = 7$ ), using the public key of user A:  $K' = (K_{AB})^{e_A} \bmod n_A = 7^3 \bmod 55 = 13$ . Please show the way in which user A determines the session key ( $K_{AB} = 7$ ), using his private key and the received enveloped key ( $K' = 13$ ).

A.  $K_{AB} = (K')^{d_A} \bmod n_A = (13)^{27} \bmod 55$

B.  $K_{AB} = (K')^{n_A} \bmod e_A = (13)^{55} \bmod 3$

C.  $K_{AB} = (K')^{e_A} \bmod n_A = (13)^3 \bmod 55$

D.  $K_{AB} = (K')^{n_A} \bmod d_A = (13)^{55} \bmod 27$

Let's break down the RSA key transport and decryption process to determine the correct formula:

### **RSA Key Transport (Simplified)**

1. **Recipient's Keys:** User A (the recipient) has a public key ( $e_A$ ,  $n_A$ ) and a private key ( $d_A$ ,  $n_A$ ). The public key is shared, the private key is kept secret.

2. **Sender's Encryption:** User B (the sender) wants to send a secret session key ( $K_{AB}$ ) to User A. User B encrypts the session key using User A's *public* key:  $K' = (K_{AB})^{e_A} \bmod n_A$  (This is what the problem provides:  $K' = 7^3 \bmod 55 = 13$ )
3. **Recipient's Decryption:** User A receives the encrypted session key ( $K'$ ). User A decrypts it using their *private* key:  $K_{AB} = (K')^{d_A} \bmod n_A$

#### Analyzing the Options:

- **A.  $K_{AB} = (K')^{d_A} \bmod n_A = (13)^{27} \bmod 55$ :** This is **correct**. It perfectly matches the decryption step in RSA. User A raises the encrypted key ( $K'$ ) to the power of their private exponent ( $d_A$ ) and takes the modulus ( $n_A$ ).
- **B.  $K_{AB} = (K')^{n_A} \bmod e_A = (13)^{55} \bmod 3$ :** This is **incorrect**. It uses the modulus and public exponent in the wrong places. It's not a valid RSA operation.
- **C.  $K_{AB} = (K')^{e_A} \bmod n_A = (13)^3 \bmod 55$ :** This is **incorrect**. This would be *re-encrypting* the encrypted key with A's public key, which would not decrypt it.
- **D.  $K_{AB} = (K')^{n_A} \bmod d_A = (13)^{55} \bmod 27$ :** This is **incorrect**. It uses the modulus and private exponent in the wrong places.

#### Verification

Let's verify option A with the given values:

- $K_{AB} = (13)^{27} \bmod 55$

Calculating this (you'd typically use a computer or calculator for modular exponentiation):

$13^{27} \bmod 55$  can be computed efficiently using repeated squaring:

$$13^1 \bmod 55 = 13$$

$$13^2 \bmod 55 = 169 \bmod 55 = 4$$

$$13^4 \bmod 55 = (13^2)^2 \bmod 55 = 4^2 \bmod 55 = 16$$

$$13^8 \bmod 55 = (13^4)^2 \bmod 55 = 16^2 \bmod 55 = 256 \bmod 55 = 36$$

$$13^{16} \bmod 55 = (13^8)^2 \bmod 55 = 36^2 \bmod 55 = 1296 \bmod 55 = 31$$

Now,  $27 = 16 + 8 + 2 + 1$

So,  $13^{27} \bmod 55 = (13^{16} * 13^8 * 13^2 * 13^1) \bmod 55$

$$= (31 * 36 * 4 * 13) \bmod 55$$

$$= (1116 * 52) \bmod 55$$

$$= (1116 \bmod 55 * 52 \bmod 55) \bmod 55$$

$$= (6 * 52) \bmod 55$$

$$= 312 \bmod 55$$

$$= 37$$

- The stated value is 7, but the correct value, according to the given parameters is 37.

**There is a typo error in the problem.** If K' was 37, then KAB would result to be 7.

Therefore, the correct formula is **A**, but, with provided values, there is a typo.

23) The Windows operating system, in order to build the certificate chain, it automatically discovers the necessary certificates in the following certificate stores:

- A. „Personal”
- B. „Intermediate Certification Authorities“
- C. „Enterprise Trust”
- D. “Trusted Root Certification Authorities”

Let's break down how Windows builds certificate chains and which certificate stores are involved:

- **Certificate Chain (or Certification Path):** This is a sequence of certificates, starting from a *trusted root CA certificate*, down through any *intermediate CA certificates*, and ending with the *end-entity certificate* (the certificate being validated). Each certificate in the chain signs the next one down, establishing a chain of trust.
- **Windows Certificate Stores:** Windows organizes certificates into logical stores. The key stores for chain building are:
  - **"Trusted Root Certification Authorities":** This is **correct** and *essential*. This store contains the self-signed certificates of root CAs that Windows trusts implicitly. The chain *must* ultimately lead back to a certificate in this store for the end-entity certificate to be considered valid.

- **"Intermediate Certification Authorities":** This is **correct** and also essential. This store contains certificates for intermediate CAs – CAs that are not root CAs but are authorized by a root CA (or another intermediate CA) to issue certificates. These certificates form the "middle" of the chain.
- **"Personal":** This store is primarily for *your* certificates (end-entity certificates issued to *you*). While Windows *can* use certificates from this store, it's not the primary place it looks for certificates to *build* the chain for validating *other* certificates. It is not generally used *automatically* for building chains during validation of others' certificates. So that option is incorrect.
- **"Enterprise Trust"** is a way to add certificates to the **Trusted Root Certification Authorities** store. It is not by itself, a root store, but an administrative method
- **Automatic Discovery Process:** When Windows needs to validate a certificate (e.g., when you connect to an HTTPS website), it tries to build a chain of trust. It looks in the "Trusted Root Certification Authorities" and "Intermediate Certification Authorities" stores to find the necessary certificates to complete the chain.

Therefore the answer is **B and D**.

24) According to the time stamping law, the time stamp information contains, among other data:

- a. the extended electronic signature of the author of the electronic data
- b. the electronic data or a hash function of it
- c. the extended electronic signature of the time-stamping issuing authority
- d. date and time

Let's examine the necessary components of a time-stamp token, according to typical time-stamping standards and laws (like RFC 3161, which is commonly referenced):

- **a. the extended electronic signature of the author of the electronic data:** This is **incorrect**. The time-stamp is issued by a Time Stamping Authority (TSA), *not* the author of the original data. The author's signature might be on the original document, but it's not part of the time-stamp token itself.

- **b. the electronic data or a hash function of it:** This is **correct**. The time-stamp doesn't usually include the *entire* original document, for efficiency and privacy. Instead, it includes a *cryptographic hash* (a one-way function) of the data. This hash acts as a unique "fingerprint" of the data. If the data changes, the hash will change, proving the data wasn't time-stamped in its current form.
- **c. the extended electronic signature of the time-stamping issuing authority:** This is **correct**. The TSA signs the time-stamp token with *its* private key. This proves the time-stamp came from the trusted TSA and hasn't been tampered with. This is what makes the time-stamp trustworthy.
- **d. date and time:** This is **correct** and the *core* purpose of the time-stamp. The time-stamp token must include a reliable, accurate date and time (usually UTC) obtained from a trusted time source. This is the time at which the TSA asserts the data (or its hash) existed.

Therefore, the correct answers are **b**, **c**, and **d**. A time-stamp token contains a hash of the data being time-stamped, a date and time, and a digital signature from the Time Stamping Authority.

25) A variation of the Diffie-Hellman scheme, for signature is:

The public elements:  $q$  (prime number),  $a < q$  (prime root of  $q-1$ )

Private key:  $X$ ,  $X < q$

Public key:  $Y = a^X \bmod q$

The message to be signed  $M$ , and its hash  $h(M)$

The signature:  $a^Z$ , where  $Z$  satisfies the equation  $Z * h = X \pmod{q-1}$

Show the way in which the verification of the signature is done:

a.  $Y = (a^Z)^M = a^X \bmod q$

b.  $Y = (a^M)^h = a^X \bmod q$

c.  $Y = (a^Z)^h = a^X \bmod q$

d.  $Y = (a^Y)^h = a^X \bmod q$

Let's analyze the provided Diffie-Hellman-like signature scheme and determine the verification process. This scheme isn't standard DSA or ECDSA; it's a variation, so we need to work through the logic based on the provided equations.

**Given:**

- Public parameters:  $q$  (prime),  $a$  (primitive root modulo  $q-1$ ). This " $q-1$ " is unusual and crucial. Standard Diffie-Hellman and DSA use a primitive root *modulo*  $q$ , not  $q-1$ .
- Private key:  $X$  (where  $X < q$ )
- Public key:  $Y = a^X \bmod q$
- Message:  $M$ , Hash of message:  $h(M)$  (we'll just call it  $h$ )
- Signature:  $a^Z$ , where  $Z * h = X \pmod{q-1}$  (This is the key equation defining the signature)

**Verification Goal:**

The verifier needs to check if the signature ( $a^Z$ ) is valid for the message ( $M$ , with hash  $h$ ) and the public key ( $Y$ ). The goal is to somehow relate  $a^Z$ ,  $h$ , and  $Y$  to confirm that the underlying private key  $X$  was used correctly.

**Deriving the Verification**

1. **Start with the signature equation:**  $Z * h = X \pmod{q-1}$
2. **Raise  $a$  to the power of both sides:**  $a^{(Z * h)} = a^X \pmod{q}$ . Remember from number theory that if  $m = n \pmod{k}$ , then  $a^m = a^n \pmod{q}$ . We will consider, as the public keys suggest  $a \bmod q$ .
3. **Use properties of exponents:**  $a^{(Z * h)} = (a^Z)^h$
4. **Substitute the public key:** We know  $Y = a^X \bmod q$ .
5. **Verification Equation:** Combining steps 3 and 4, the verification equation becomes:  $(a^Z)^h = Y \pmod{q}$
6. We should compare the left and the right member

**Analyzing the Options:**

- **a.  $Y = (a^Z)^M = a^X \bmod q$ :** Incorrect. Uses the message  $M$  directly, not the hash  $h$ .
- **b.  $Y = (a^M)^h = a^X \bmod q$ :** Incorrect. Uses the message  $M$  in a way that doesn't relate to the signature equation.

- **c.  $Y = (a^Z)^h = a^X \bmod q$ :** This is **correct**. It exactly matches the derived verification equation. The verifier computes  $(a^Z)^h \bmod q$  using the signature  $(a^Z)$ , the hash of the message ( $h$ ), and the public parameter  $q$ . If the result equals the sender's public key  $Y$ , the signature is valid.
- **d.  $Y = (a^Y)^h = a^X \bmod q$ :** Incorrect. Uses the public key  $Y$  within the exponentiation, which doesn't make sense in this context.

Therefore, the correct answer is **c**.

26) In Chaum blind signature scheme for (blinding) A computes:  $m^* = mk^e \bmod n$ . Indicate the signing and un-blinding mechanism:

- A. (un-blinding) A computes:  $s^* = (m^*)^d \bmod n$
- B. (signing) B computes:  $s = k^{(-1)}s^* \bmod n$
- C. (un-blinding) A computes:  $s = k^{(-1)}s^* \bmod n$
- D. (signing) B computes:  $s^* = (m^*)^d \bmod n$

Let's break down the Chaum blind signature scheme and analyze the signing and unblinding steps:

### Chaum Blind Signature Overview

The goal of a blind signature is for user A to get a message  $m$  signed by user B *without* B learning the content of  $m$ . This is achieved through a blinding factor.

#### Steps (Standard Chaum Scheme):

##### 1. Blinding (by A):

- A chooses a random blinding factor  $k$ , where  $k$  is relatively prime to  $n$  (the RSA modulus of B). This means  $\gcd(k, n) = 1$ .
- A computes the blinded message  $m^* = m \cdot k^e \bmod n$ , where  $e$  is B's public exponent. This is what's given in the problem.

##### 2. Signing (by B):

- B receives  $m^*$  (the blinded message).
- B signs  $m^*$  using their private key  $d$ , computing  $s^* = (m^*)^d \bmod n$ . B *doesn't* know  $m$ .

##### 3. Unblinding (by A):

- A receives  $s^*$  (the signed blinded message).

- A removes the blinding factor to obtain the signature  $s$  on the original message  $m$ :  $s = k^{(-1)} * s^* \bmod n$ . This works because  $k^{(-1)}$  is the modular inverse of  $k$  modulo  $n$ .

### Analyzing the Options:

- *A. (un-blinding) A computes:  $s = (m)^d \bmod n$ :* This is **incorrect**. This describes the *signing* step performed by B, not the unblinding step performed by A.
- *B. (signing) B computes:  $s = k^{(-1)}s^* \bmod n$ :* This is **incorrect**. This describes the *unblinding* step performed by A, not the signing step performed by B. The roles are reversed.
- *C. (un-blinding) A computes:  $s = k^{(-1)}s^* \bmod n$ :* This is **correct**. This is the correct unblinding step. A multiplies the signed blinded message ( $s^*$ ) by the modular inverse of the blinding factor ( $k^{(-1)}$ ) to obtain the signature ( $s$ ) on the original message.
- *D. (signing) B computes:  $s = (m)^d \bmod n$ :* This is **correct**. This is the correct signing step. B signs the blinded message ( $m^*$ ) using their private key ( $d$ ) to produce the signed blinded message ( $s^*$ ).

Therefore, the correct answers are **C** and **D**. They correctly describe the unblinding and signing steps, respectively, in the Chaum blind signature scheme.

27) The correct equation, which has to be verified for validating the signature, is:

In Shamir's identity based signature scheme, in which  $(e, d)$  are the RSA master keys, the signing  $(s, t)$  has the following steps:

- it randomly chooses  $r$ ;
- it computes  $t = r^e \bmod n$ ;
- it computes  $f = H(t, m)$ ;
- it computes  $sID = ID^d \bmod n$ ;
- it computes  $s = sID * r^f \bmod n$ .

A.  $s^e = ? = ID * t^H(t, m) \bmod n$

B.  $s^e = ? = H(t, m) * t^ID \bmod n$

C.  $s^e = ? = H(m) * t^ID \bmod n$

D.  $s^e = ? = t^H(t, m) \bmod n$

Let's break down Shamir's identity-based signature scheme and derive the verification equation. This is *not* standard RSA or DSA; it's a different scheme based on identities.

### Given Information:

- RSA master keys:  $(e, d)$  (where  $e$  is public,  $d$  is private, and  $n$  is the modulus – we assume a standard RSA setup where  $e \cdot d = 1 \bmod \phi(n)$ )
- Signing steps:
  1. Choose random  $r$ .
  2. Compute  $t = r^e \bmod n$ .
  3. Compute  $f = H(t, m)$  (where  $H$  is a hash function, and  $m$  is the message).
  4. Compute  $sID = ID^d \bmod n$  (where  $ID$  is the user's identity, treated as a number).
  5. Compute  $s = sID * r^f \bmod n$ .
- Signature:  $(s, t)$

### Verification Goal:

The verifier has the message  $m$ , the signature  $(s, t)$ , the user's identity  $ID$ , and the public master key  $e$  and modulus  $n$ . They need to verify if the signature is valid.

### Derivation:

1. **Start with the signature  $s$ :**  $s = sID * r^f \bmod n$
2. **Substitute  $sID$ :**  $s = (ID^d) * r^f \bmod n$
3. **Raise both sides to the power of  $e$ :**  $s^e = ((ID^d) * r^f)^e \bmod n$
4. **Use properties of exponents and modular arithmetic:**  $s^e = (ID^d)^e * (r^f)^e \bmod n = ID^{d \cdot e} * (r^e)^f \bmod n$
5. **Since  $d$  and  $e$  are RSA keys,  $d \cdot e = 1 \bmod \phi(n)$ :** This means  $ID^{d \cdot e} = ID^1 = ID \bmod n$ . This is a crucial property of RSA.
6. **Substitute  $t = r^e \bmod n$ :**  $s^e = ID * t^f \bmod n$
7. **Substitute  $f = H(t, m)$ :**  $s^e = ID * t^{H(t, m)} \bmod n$
8. **Verification Equation:** The final verification equation is:  $s^e = ID * t^{H(t, m)} \bmod n$

### Analyzing the Options:

- **A.  $s^e = ? = ID * t^{H(t, m)} \bmod n$ :** This is **correct**. It exactly matches our derived verification equation.

- **B.  $s^e = ? = H(t, m) * t^{ID} \bmod n$ :** Incorrect. The roles of ID and  $H(t, m)$  are swapped, and the exponentiation is wrong.
- **C.  $s^e = ? = H(m) * t^{ID} \bmod n$ :** Incorrect. Uses  $H(m)$  instead of  $H(t, m)$ , and the exponentiation is wrong.
- **D.  $s^e = ? = t^{H(t, m)} \bmod n$ :** Incorrect. It's missing the ID component, which is essential for identity-based signatures.

Therefore, the correct answer is **A**. The verifier computes  $s^e \bmod n$  and  $ID * t^{H(t, m)} \bmod n$  and checks if they are equal. If they are, the signature is valid.

28) About blind signatures, we can say that:

- a. they were theoretically developed by Chaum
- b. they were theoretically developed by Chaum-van Antwerpen
- c. they were initially designed as a protocol for securing web apps
- d. they are used in the field of electronic cash

Let's evaluate the statements about blind signatures:

- **a. they were theoretically developed by Chaum:** This is **correct**. David Chaum introduced the concept of blind signatures in his 1982 paper, "Blind Signatures for Untraceable Payments." This is the foundational work on the topic.
- **b. they were theoretically developed by Chaum-van Antwerpen:** This is **incorrect**. While van Antwerpen made contributions to cryptography, blind signatures are primarily attributed to David Chaum.
- **c. they were initially designed as a protocol for securing web apps:** This is **incorrect**. Blind signatures were initially conceived for *untraceable electronic payments*, not general web application security. The original motivation was to create a digital equivalent of cash, where the payer's identity could be protected during a transaction.
- **d. they are used in the field of electronic cash:** This is **correct**. This is the primary application for which blind signatures were originally developed and are still most relevant. They allow for anonymous digital cash systems.

Therefore, the correct answers are **a** and **d**. David Chaum developed the concept, and a key application is in electronic cash systems.

29) MQV (Menezes-Qu-Vanstone) is:

- a. an authenticated protocol for establishing a key, based on Diffie-Hellman
- b. an authentication protocol based on Diffie-Hellman
- c. based on a pair of keys: a static one and an ephemeral one
- d. a distribution key protocol, based on Diffie-Hellman

Let's analyze the properties of the MQV (Menezes-Qu-Vanstone) protocol:

- **MQV Overview:** MQV is a key agreement protocol, meaning it allows two parties to establish a shared secret key over an insecure channel. It's known for its efficiency and security properties. It builds upon the Diffie-Hellman protocol but adds features to prevent certain attacks.
- **a. an authenticated protocol for establishing a key, based on Diffie-Hellman:** This is **correct**. MQV is a key agreement protocol (establishes a shared key) and it's *authenticated*, meaning the parties can be sure of each other's identities. It builds upon the basic Diffie-Hellman exchange.
- **b. an authentication protocol based on Diffie-Hellman:** This is **partially correct, but incomplete**. MQV is *more* than just authentication; it's primarily about key *agreement*. While it provides authentication as part of the process, its main goal is to establish a shared secret key. So, while not *wrong*, it's not the best description.
- **c. based on a pair of keys: a static one and an ephemeral one:** This is **correct**. A key feature of MQV is that it uses *both* long-term (static) private/public key pairs *and* short-term (ephemeral) private/public key pairs. This combination provides important security properties, including known-key security and (with proper implementation) forward secrecy.
- **d. a distribution key protocol, based on Diffie-Hellman:** This is **partially correct, but "key agreement" is more precise**. MQV isn't about *distributing* a pre-existing key; it's about two parties *collaboratively deriving* a shared secret key. So, while it's related to key distribution, "key agreement" is the more accurate term.

Therefore, the best answers are **a** and **c**. "a" gives a good overall description, and "c" highlights a key architectural feature of MQV. "b" and "d" are partially correct but less precise than "a".

30) DSA is a cryptographic algorithm based on:

- a. the difficulty of multiplying large prime numbers
- b. the difficulty of computing exponents in finite field
- c. the difficulty of large integer numbers factorization
- d. the difficulty of computing discrete logarithms in finite field

Let's examine the mathematical basis of the Digital Signature Algorithm (DSA):

- **a. the difficulty of multiplying large prime numbers:** This is incorrect. Multiplying large prime numbers is *easy*. The difficulty lies in the *reverse* operation (factoring). This describes the basis of RSA, not DSA.
- **b. the difficulty of computing exponents in finite field:** This is incorrect. Exponentiation in a finite field is computationally *efficient*. DSA's security relies on the difficulty of the *inverse* operation.
- **c. the difficulty of large integer numbers factorization:** This describes the basis of RSA, *not* DSA.
- **d. the difficulty of computing discrete logarithms in finite field:** This is **correct**. DSA's security rests on the Discrete Logarithm Problem (DLP) in a finite field. Given  $g$ ,  $y$ , and  $p$ , where  $y = g^x \bmod p$ , it's computationally very difficult to find  $x$  (the discrete logarithm of  $y$  to the base  $g$  modulo  $p$ ).

Therefore, the correct answer is **d**.

31) Electronic signatures are based on:

- a. scanned handwritten signatures
- b. cryptographic systems with public keys and hash functions
- c. biometric data (finger print, iris, etc.)
- d. cryptographic systems with secret keys and hash functions

Let's clarify what constitutes a true *electronic signature* (often used interchangeably with *digital signature* in a technical context) and distinguish it from related concepts:

- **a. scanned handwritten signatures:** This is **incorrect**. A scanned handwritten signature is simply a *digital image* of a handwritten signature. It has *no inherent cryptographic security*. It can be easily copied and pasted, providing no real

authentication or integrity. This is *digitized*, but not a *digital* signature in the cryptographic sense.

- **b. cryptographic systems with public keys and hash functions:** This is **correct**. This describes the foundation of digital signatures. A public-key cryptosystem (like RSA or ECDSA) is used, where the signer has a private key (for signing) and a corresponding public key (for verification). Hash functions are used to create a unique "fingerprint" of the document being signed.
- **c. biometric data (fingerprint, iris, etc.):** This is **incorrect**. Biometric data is used for *authentication* (verifying a person's identity), but it's not directly used to *create* a digital signature. Biometrics might be used as part of a *multi-factor authentication* system to *unlock* a device that *then* uses a private key to sign, but the biometric data itself isn't the signature.
- **d. cryptographic systems with secret keys and hash functions:** This is describing *symmetric* cryptography, which is different than public-key signature schemes. While Symmetric cryptography and hash functions are often used together (e.g. HMAC), they are not sufficient for a digital signature scheme as defined in the industry. So, this is **incorrect**

Therefore, the correct answer is **b**. True digital signatures rely on public-key cryptography and hash functions to provide authentication, integrity, and non-repudiation.

32) Signing a document is done using:

- a. the private key of the private message sender
- b. the private key of the message recipient
- c. the sender's public key message
- d. the public key of the recipient

Let's clarify who uses which key in the digital signature process:

- **Digital Signature Process:**
  1. **Signing:** The *sender* uses their *private* key to create a digital signature for a message.
  2. **Verification:** The *recipient* (or anyone) uses the sender's *public* key to verify the signature.
- **Analyzing the Options:**

- **a. the private key of the private message sender:** This is **correct**. The sender *always* signs with their own *private* key. This is fundamental to digital signatures.
- **b. the private key of the message recipient:** This is **incorrect**. The recipient's private key is never used in the signing process. The recipient uses the *sender's public key* for *verification*.
- **c. the sender's public key message:** This is **incorrect**. The sender's public key is used for *verification*, not signing.
- **d. the public key of the recipient:** This is **incorrect**. The recipient's public key is not involved in either signing or verification of a digital signature. The recipient's keys would be used if the sender were *encrypting* the message for confidentiality, but that's a separate operation from signing.

Therefore, the correct answer is **a**. The sender *always* uses their own *private* key to create a digital signature.

33) Signature Verification of a document is done using:

- a. the private key of message sender
- b. the private key of the message recipient
- c. the public key of the sender
- d. the public key of the recipient

Let's review the key used for verifying a digital signature:

- **Verification Process:** To verify a digital signature, you need the *public key* of the entity that supposedly created the signature (the sender). You use this public key, along with the signature itself and the original message (or its hash), to check if the signature is valid.
- **Analyzing the Options:**
  - **a. the private key of message sender:** This is **incorrect**. The sender's private key is used for *creating* the signature, not verifying it. Private keys are kept secret.
  - **b. the private key of the message recipient:** This is **incorrect**. The recipient's private key is not involved in signature verification.
  - **c. the public key of the sender:** This is **correct**. This is the fundamental principle of signature verification. The sender's *public* key is used to verify a signature created with the corresponding *private* key.

- **d. the public key of the recipient:** This is **incorrect**. The recipient's public key is not used in signature verification.

Therefore, the correct answer is **c**. You *always* use the *sender's public key* to verify a digital signature.

34) For electronic signature can be used the next cryptographic algorithms:

- a. RSA, DSA, DSA over elliptic curves
- b. SHA-1
- c. AES over elliptic curve
- d. DES

Let's distinguish between algorithms suitable for *digital signatures* and other cryptographic algorithms:

- **Digital Signature Algorithms:** These algorithms are specifically designed to create and verify digital signatures, providing authentication, integrity, and non-repudiation. They are *asymmetric* (public-key) algorithms.
- **Hashing Algorithms:** These algorithms create a one-way "fingerprint" of data. They are used *in conjunction with* signature algorithms, but they don't create signatures themselves.
- **Symmetric Encryption Algorithms:** These algorithms use the *same* key for encryption and decryption. They provide confidentiality, but not authentication or non-repudiation in the way signature algorithms do.
- **Analyzing the Options:**
  - **a. RSA, DSA, DSA over elliptic curves:** This is **correct**.
    - RSA: Can be used for both encryption and digital signatures.
    - DSA: Specifically designed for digital signatures.
    - DSA over elliptic curves (ECDSA): A variant of DSA that uses elliptic curve cryptography, offering smaller key sizes for equivalent security.
  - **b. SHA-1:** This is **incorrect**. SHA-1 is a *hashing algorithm*, not a signature algorithm. It's used to create a hash of the message, which is then signed, but SHA-1 itself doesn't perform the signing. Also, SHA-1 is considered cryptographically broken and should not be used.

- **c. AES over elliptic curve:** This is **incorrect**. AES (Advanced Encryption Standard) is a *symmetric encryption algorithm*. "AES over elliptic curve" doesn't make sense. Elliptic curve cryptography is used for *asymmetric* algorithms (like ECDSA and ECDH). There is no such thing.
- **d. DES:** This is **incorrect**. DES (Data Encryption Standard) is a *symmetric encryption algorithm* (and an outdated one at that). It's not used for digital signatures.

Therefore, the correct answer is **a**. RSA, DSA, and ECDSA (DSA over elliptic curves) are all established algorithms for creating digital signatures.

35) XML signatures are:

- a. data structures containing the signature value and signature algorithms used
- b. data structures containing the signature value and the necessary data for the verification process
- c. at the enveloped XML signatures, the signed object encapsulates signature
- d. at the enveloped XML signatures, the signed object encapsulates signature

Let's break down XML Signatures and their different types:

- **XML Signature Basics:** XML Signatures provide a standard way to digitally sign XML documents (or parts of them). They ensure data integrity, message authentication, and (optionally) non-repudiation. The signature information is itself represented in XML.
- **Key Components:** An XML Signature includes:
  - **Signature Value:** The actual cryptographic signature, calculated over the signed data.
  - **SignedInfo:** Information about *what* was signed (references to the data), the algorithms used, and canonicalization methods (to ensure consistent processing).
  - **KeyInfo (Optional):** Information about the key used to create the signature (e.g., a certificate). This can help the verifier locate the necessary public key.
  - **Other Data**
- **Types of XML Signatures (Placement of Signature Element):**

- **Enveloped Signature:** The signature element is *inside* the XML document being signed. The signature becomes part of the signed content.
- **Enveloping Signature:** The signature element *contains* the XML document being signed. The signed content is wrapped *within* the signature.
- **Detached Signature:** The signature element and the signed XML document are *separate*. The signature references the external document.
- **Analyzing the Options:**
  - **a. data structures containing the signature value and signature algorithms used:** This is **partially correct, but incomplete**. XML Signatures contain the signature value and information about the algorithms *used*, but they also contain *references* to the data that was signed (in the SignedInfo element) and optionally information about the key (in the KeyInfo element).
  - **b. data structures containing the signature value and the necessary data for the verification process:** This is **correct** and a more complete description. An XML Signature includes the signature value, information about the signed data and algorithms (in SignedInfo), and often information to help locate the verification key (in KeyInfo). All this information is needed for verification.
  - **c. at the enveloped XML signatures, the signed object encapsulates signature:** Typo here ("enveloped" should be "enveloped"). This describes an *Enveloping* signature, so it's mis-categorized, and therefore is **incorrect**. The question has a typo, as well, mixing 2 words.
  - **d. at the enveloped XML signatures, the signed object encapsulates signature:** There is a typo, it should read: *at the enveloping signatures, the signature encapsulates the signed object*. This would be **correct**. This statement has another typo, as it describes Enveloping signature, and it says 'enveloped'. So, the statement is **incorrect**

Therefore, the best answer is **b**. It provides the most accurate and complete description of the contents of an XML Signature. Option 'a' is incomplete. Options 'c' and 'd' have typos and mix up the definitions of Enveloping and Enveloped signature placement.

### 36) Time stamping

a. it is realized by the the application and countersigned by a time stamping service provider

- b. it is legal regulated by time stamping law enacted in Romania in 2004
- c. it is legal regulated by RFC-3161 Internet X.509 PKI Time Stamp Protocol (TSP);
- d. is performed by a certification service provider

Let's revisit time-stamping and analyze the given options:

- **a. it is realized by the application and countersigned by a time-stamping service provider:** This is **correct**. This accurately describes the typical time-stamping process:
  1. The application (or user) generates a hash of the data to be time-stamped.
  2. This hash is sent to the Time Stamping Authority (TSA).
  3. The TSA combines the hash with the current time from a trusted source and digitally signs the result (this is the "countersigning"). This creates the time-stamp token.
- **b. it is legal regulated by time stamping law enacted in Romania in 2004:** This is **likely correct**, as many countries adopted laws around that time to address digital signatures and time-stamping. Without knowing the specific Romanian law number, we can't be 100% certain, but it's a very reasonable assumption.
- **c. it is legal regulated by RFC-3161 Internet X.509 PKI Time Stamp Protocol (TSP):** This is **partially correct, but important nuance**. RFC 3161 is a *technical standard*, not a *law*. It defines the *protocol* for how time-stamping services should operate. Laws *often reference* technical standards like RFC 3161 to provide specifics, but the RFC itself isn't a law. So, it's the *technical* standard, not the legal one.
- **d. is performed by a certification service provider:** This is **partially correct, but imprecise**. The entity providing the time-stamping service is more accurately called a *Time Stamping Authority (TSA)*. A Certificate Authority (CA) *can* also be a TSA, but the roles are distinct. It's better to use the specific term "Time Stamping Authority" or "Time-Stamping Service Provider."

Therefore, the best answers are **a** and **b**. "a" describes the process accurately, and "b" is a reasonable assumption about legal regulation. "c" describes the *technical* standard, not the *legal* regulation, and "d" uses imprecise terminology.

37) RFC 2587 (Internet X.509 PKI Operational Protocols - LDAPv2):

- a. defines the access method to the certificates repository

b. is a PKCS standard developed by RSA Data Security for access to storage repository

c. defines the attributes and object classes supported by the LDAP server

d. defines the method of online checking certificates

Let's break down RFC 2587 and its role in PKI:

- **RFC 2587's Focus:** This RFC specifically deals with using LDAP version 2 (LDAPv2) as a protocol for accessing information within a Public Key Infrastructure (PKI). This information primarily includes certificates and Certificate Revocation Lists (CRLs).
- **Analyzing the Options:**
  - **a. defines the access method to the certificates repository:** This is **correct**. RFC 2587 outlines how LDAPv2 can be used to *query* and *retrieve* certificates and CRLs from a directory server. It defines how a client can connect to the server, search for certificates, and retrieve them.
  - **b. is a PKCS standard developed by RSA Data Security for access to storage repository:** This is **incorrect**. RFC 2587 is an *IETF* (Internet Engineering Task Force) standard, not a PKCS (Public-Key Cryptography Standards) standard developed by RSA Data Security. PKCS standards have different numbers (e.g., PKCS#7, PKCS#12).
  - **c. defines the attributes and object classes supported by the LDAP server:** This is **correct**. A crucial part of RFC 2587 is defining the *schema* for storing PKI information within an LDAP directory. This includes defining the object classes (like certificate, cRLDistributionPoint, etc.) and the attributes within those classes (like userCertificate, certificateRevocationList, etc.) that hold the relevant data.
  - **d. defines the method of online checking certificates:** This is **incorrect**. RFC 2587 defines how to *access* certificate and CRL information via LDAP. *Online* certificate checking is typically done using the Online Certificate Status Protocol (OCSP), defined in RFC 6960 (and earlier versions). While an LDAP server *could* be used as a source of data for an OCSP responder, RFC 2587 itself doesn't define the OCSP protocol.

Therefore, the correct answers are **a** and **c**. RFC 2587 defines how to use LDAPv2 to access PKI information (certificates and CRLs) and specifies the schema (attributes and object classes) for storing that information.

38) For an RSA cryptographic system with public key ( $e = 5$ ,  $n = 21$ ), the pair private key ( $d$ ,  $n$ ) of

this key is:

Let's determine the private key for the given RSA public key.

### **RSA Key Generation (Simplified)**

1. **Choose two distinct prime numbers,  $p$  and  $q$ .**
2. **Calculate  $n = p * q$ .**  $n$  is the modulus.
3. **Calculate  $\phi(n) = (p-1) * (q-1)$ .**  $\phi(n)$  is Euler's totient function, representing the number of integers less than  $n$  that are relatively prime to  $n$ .
4. **Choose an integer  $e$  (the public exponent) such that  $1 < e < \phi(n)$  and  $\gcd(e, \phi(n)) = 1$**  (i.e.,  $e$  and  $\phi(n)$  are coprime).
5. **Calculate  $d$  (the private exponent) such that  $d * e \equiv 1 \pmod{\phi(n)}$ .**  $d$  is the modular multiplicative inverse of  $e$  modulo  $\phi(n)$ .

**Given:**

- Public key: ( $e = 5$ ,  $n = 21$ )

### **Finding the Private Key ( $d$ )**

1. **Find  $p$  and  $q$ :** Since  $n = 21$ , and  $n = p * q$ , we can easily factor 21:  $p = 3$  and  $q = 7$  (or vice versa).
2. **Calculate  $\phi(n)$ :**  $\phi(n) = (p-1) * (q-1) = (3-1) * (7-1) = 2 * 6 = 12$
3. **Find  $d$ :** We need to find  $d$  such that  $5 * d \equiv 1 \pmod{12}$ . In other words,  $5 * d$  should leave a remainder of 1 when divided by 12. We can try values for  $d$ :
  - $d = 1$ :  $5 * 1 \pmod{12} = 5$
  - $d = 2$ :  $5 * 2 \pmod{12} = 10$
  - $d = 3$ :  $5 * 3 \pmod{12} = 3$
  - $d = 4$ :  $5 * 4 \pmod{12} = 8$
  - $d = 5$ :  $5 * 5 \pmod{12} = 1$  <- This is it!
4. **Private Key:** Therefore,  $d = 5$ . The private key is ( $d = 5$ ,  $n = 21$ ).

### **The Answer**

The private key pair is ( $d=5$ ,  $n=21$ ).

39) Among static biometric methods are:

- a. fingerprint
- b. Hand geometry
- c. identification of the retina
- d. Voice recognition

Let's categorize the given biometric methods as static or dynamic:

- **Static Biometrics:** These methods rely on physical characteristics that are relatively stable and don't change significantly over time (at least in adulthood). They are typically measured at a single point in time.
- **Dynamic Biometrics:** These methods rely on behavioral characteristics or physiological processes that change over time. They require measuring a *pattern* of activity over a period.
- **Analyzing the Options:**
  - **a. fingerprint:** This is **static**. Fingerprint patterns are generally fixed from birth (barring injury).
  - **b. Hand geometry:** This is **static**. The shape and size of the hand (including finger length, width, etc.) are relatively stable physical characteristics.
  - **c. identification of the retina:** This is **static**. The pattern of blood vessels in the retina is unique and generally stable throughout life.
  - **d. Voice recognition:** This is **dynamic**. Voice recognition relies on the *way* a person speaks, including intonation, speed, accent, and other dynamic characteristics. These can vary depending on factors like mood, health, and environment. While the underlying vocal tract structure is relatively static, the *way* it's used is dynamic.

Therefore, the correct answers are **a**, **b**, and **c**. Fingerprint, hand geometry, and retinal patterns are static biometric traits. Voice recognition is a dynamic biometric trait.

40) Main current issues concerning the use of biometric signature methods are:

- a. the reliability of biometric identification methods
- b. lack of appropriate standard
- c. lack of smartcards with biometric sensors

d. the impossibility of memorizing biometric characteristics due to lack of space on card

Let's examine the current challenges and issues surrounding the use of biometric signature methods (specifically, *dynamic* signature recognition, where the *way* you sign is measured, not just a static image of your signature):

- **a. the reliability of biometric identification methods:** This is **correct** and a major concern. Biometric methods, including dynamic signature recognition, are not perfect. Factors like variability in signing style (due to mood, illness, writing surface, etc.), sensor accuracy, and algorithm performance can all affect reliability. False positives (accepting an invalid signature) and false negatives (rejecting a valid signature) are possibilities.
- **b. lack of appropriate standard:** This is **correct** and another significant hurdle. While there are *some* standards related to biometric data formats and interoperability (e.g., ISO/IEC 19794), there isn't a single, universally accepted standard specifically for dynamic signature recognition that covers all aspects (data format, performance metrics, security requirements, etc.). This lack of standardization can lead to compatibility issues and difficulties in comparing different systems.
- **c. lack of smartcards with biometric sensors:** This is **incorrect** and outdated. Smart cards with integrated fingerprint sensors *do* exist and are used for various applications, including secure access and authentication. While not *ubiquitous*, the technology is readily available. This isn't a primary limiting factor for *dynamic signature* recognition specifically.
- **d. the impossibility of memorizing biometric characteristics due to lack of space on card:** This is **incorrect** and misunderstands how biometric data is used with smart cards. The *biometric template* (a mathematical representation of the biometric trait) *can* be stored on a smart card. The card doesn't "memorize" in the human sense; it *stores* the template. The size of a fingerprint or dynamic signature template is relatively small, and smart cards have sufficient storage for this. Furthermore, "match-on-card" technology exists, where the biometric comparison happens *inside* the secure chip of the smart card, so the template never leaves the card. The problem talks about memorizing, not storing.

Therefore, the correct answers are **a** and **b**. Reliability of the biometric methods and the lack of comprehensive, widely adopted standards are significant challenges for dynamic signature recognition. Options c and d describe problems that do not exist.

41) The electronic signatures Directive issued by the Commission of the European Communities the subject of debate and adoption in the Parliament and the European Council in 1999 refers to:

- a. the definition of terminology used in electronic signature
- b. the legal effects of electronic signatures
- c. international aspects of using electronic signatures
- d. the requirements for certification service providers and how their certification and authorization

Let's analyze what the 1999 EU Electronic Signatures Directive (Directive 1999/93/EC) covered. This directive was a landmark piece of legislation that established a legal framework for electronic signatures within the European Union.

- **a. the definition of terminology used in electronic signature:** This is **correct**. The Directive defined key terms like "electronic signature," "advanced electronic signature," "signatory," "signature creation data," "certificate," etc. This was crucial for establishing a common understanding across member states.
- **b. the legal effects of electronic signatures:** This is **correct**. The Directive's primary goal was to ensure that electronic signatures would be legally recognized and admissible as evidence in legal proceedings. It established the principle of non-discrimination – that an electronic signature couldn't be denied legal effect solely because it was in electronic form. It also distinguished between different *types* of electronic signatures and their respective legal effects.
- **c. international aspects of using electronic signatures:** This is **partially correct**. While the Directive's primary focus was on the *internal* EU market, it did include provisions related to certificates issued by certification service providers established *outside* the EU. It aimed to facilitate cross-border recognition, but this was a more limited aspect compared to the internal market focus.
- **d. the requirements for certification service providers and how their certification and authorization:** This is **correct**. The Directive laid down requirements for Certification Service Providers (CSPs), particularly those issuing "qualified certificates" (which were linked to the highest level of assurance for electronic signatures). It covered aspects like security requirements, liability, and data protection. It *didn't* mandate a specific accreditation/authorization scheme, but it set out criteria that member states could use to establish such schemes.

Therefore, all the answers (**a**, **b**, **c**, and **d**) are correct. The 1999/93/EC Directive was comprehensive, covering definitions, legal effects, some international aspects, and requirements for CSPs. It's important to note this Directive has since been replaced by

the eIDAS Regulation (Regulation (EU) No 910/2014), which builds upon and expands the framework established by the 1999 Directive.

42) A secure mechanism to create electronic signature must meet the following conditions:

a. not to change the content of the document that must be electronically signed and shall not prevent it to be presented to signer prior to the completion of signature

b. signature creation data can not be deducted and the signature to be protected against

forgery

c. signature creation data can not occur only once and their confidentiality to be assured

d. signature creation data to be effectively protected by the signatory against their use by unauthorized persons

This question repeats a previous one. Let's go through the options for creating a *secure* electronic signature:

- **a. not to change the content of the document that must be electronically signed and shall not prevent it to be presented to signer prior to the completion of signature:** This is **correct** and reflects the WYSIWYG (What You See Is What You Get) principle. The signing process shouldn't alter the document, and the signer *must* be able to see and review the document *before* signing.
- **b. signature creation data can not be deducted and the signature to be protected against forgery:** There is a typo in the question, that should read: *signature creation data can not be deduced....* This is **correct**. The *private key* (the signature creation data) must be computationally infeasible to derive from the public key or any other public information. The entire signature scheme must be resistant to forgery – it should be impossible for someone to create a valid signature without the private key.
- **c. signature creation data can not occur only once and their confidentiality to be assured:** It contains a typo. It should read "data" instead of "dates". The statement would mean that the private key must be kept confidential. Also, the "can not occur only once" is technically wrong, because the private key *must* be created once. So this option is **incorrect**.
- **d. signature creation data to be effectively protected by the signatory against their use by unauthorized persons:** It contains a typo. It should read

"data" instead of "dates". This is **correct**. The signer *must* protect their private key (the signature creation data). If the private key is compromised, the security of the signature is lost.

Therefore, **a, b and d** are correct. Option c is not correct due to the typo, and the requirement of being created only once.

43) Accreditation of certification service providers:

- a. is mandatory
- b. is done by the Regulatory Authority and Supervision
- c. is voluntary
- d. regularly made by surveillance procedure

Let's analyze the statements about the accreditation of Certification Service Providers (CSPs):

- **Accreditation Context:** Accreditation, in this context, refers to a formal process where an independent body assesses a CSP's operations, security, and procedures to ensure they meet specific standards (like those for issuing qualified certificates under eIDAS or similar regulations).
- **a. is mandatory:** This is **generally incorrect, but depends on the context**. Accreditation is *not always mandatory* for *all* CSPs. A CSP can issue certificates *without* being accredited. However, accreditation (or a similar process like being designated a "qualified trust service provider" under eIDAS) is often *required* to issue certificates that have the *highest level of legal recognition* (e.g., "qualified certificates" in the EU). So, it's mandatory for *certain types* of certificates and services, but not universally for all CSPs.
- **b. is done by the Regulatory Authority and Supervision:** This is **generally correct**. Accreditation is typically performed by a designated national accreditation body or a supervisory authority that has the expertise and authority to assess CSPs against the relevant standards. The specific body varies by country.
- **c. is voluntary:** This is **partially correct, but misleading**. As explained in point (a), accreditation is *voluntary* in the sense that a CSP doesn't *have* to be accredited to operate. However, it's often *required* to provide certain types of services (like issuing qualified certificates). So, while technically voluntary, it's often practically necessary for certain business models.

- **d. regularly made by surveillance procedure:** This is **correct**. Accreditation isn't a one-time event. Accredited CSPs are typically subject to ongoing surveillance and regular audits to ensure they continue to meet the required standards. This includes periodic reviews, audits, and potentially even unannounced inspections.

Therefore, the best answers, considering the nuances, are **b** and **d**. Accreditation is typically performed by a designated authority and involves ongoing surveillance. "a" and "c" are partially correct but need the crucial context that accreditation is often *required* for specific types of certificates/services, even if not universally mandatory for all CSPs. The best choice between them is c, because it highlights a CSP *can* operate without accreditation.

44) DSA is a cryptographic algorithm based on:

- a. the difficulty of computing discrete logarithms in finite fields
- b. the difficulty of large integer numbers factorization
- c. the difficulty of computing exponents in finite field
- d. the difficulty of multiplying large prime numbers

This is a repeat question. Let's reiterate the correct answer and explanation for DSA (Digital Signature Algorithm):

- **a. the difficulty of computing discrete logarithms in finite fields:** This is **correct**. DSA's security is based on the Discrete Logarithm Problem (DLP) in a finite field. Given  $g$ ,  $y$ , and  $p$ , where  $y = g^x \bmod p$ , it's computationally infeasible to find  $x$  (the discrete logarithm).
- **b. the difficulty of large integer numbers factorization:** This is the basis of RSA, *not* DSA.
- **c. the difficulty of computing exponents in finite field:** Exponentiation is *easy*. DSA relies on the difficulty of the *inverse* operation (the discrete logarithm).
- **d. the difficulty of multiplying large prime numbers:** Multiplying primes is easy. Factoring the result back into the primes is hard (and the basis of RSA).

Therefore, the correct answer is **a**.

